

- (1) Write the Matlab or Python code to compute the FFT of two N-point real signals x and y using only one N-point FFT. (20 scores)

[Fx, Fy] = fftreal(x, y)

The code should be handed out by NTUCool.

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Input real sequence f1[n]:
[1, 2, 3, 4]
Input real sequence f2[n]:
[4, 8, 7, 3]
DFT Result F1:
[10.+0.j -2.+2.j -2.+0.j -2.-2.j]
DFT Result F2:
[22.-0.j -3.-5.j -0.+0.j -3.+5.j]
```

- (2) Compared to the original non-sectioned convolution, what are the two main advantages of the sectioned convolution? (7 scores)

1. 降低計算複雜度. non-sectioned convolution: $O(N \log N)$
sectioned convolution: $O(N)$

2. 在實際上, 如果傅立葉轉換器數過多, 硬體無法因應大量不同長度的傅立葉轉換, 硬體成本過大

- (3) Design an efficient way to compute the convolution operation of $y[n] = x[n] * h[n]$ where $h[n] = [0.09 \ 0.36 \ 0.55 \ 0.55 \ 0.36 \ 0.09]$ $-2 \leq n \leq 3$

The number of multiplications should be as small as possible. (10 scores)

$$y[n] = 0.09 x[n-2] + 0.36 x[n-1] + 0.55 x[n] + 0.55 x[n+1] + 0.36 x[n+2] + 0.09 x[n+3]$$

$$= 0.09 (x[n-2] + 4x[n-1] + 9x[n+2] + x[n+3]) + 0.55 (x[n] + x[n+1])$$

if $x[n]$ is real and $\text{length}(x) = N$, $\approx N$ MULs.

M 和 N 差很多 $\rightarrow$ direct.

(4) Suppose that $\text{length}(x[n]) = 1500$. What is the best way to implement the convolution of $x[n]$ and $y[n]$ if

(a) $\text{length}(y[n]) = 250$, (b) $\text{length}(y[n]) = 50$, (c) $\text{length}(y[n]) = 10$, (d) $\text{length}(y[n]) = 2$?

Please show

(i) the calculation method (direct, non-sectioned convolution, or sectioned convolution),

(ii) the number of points of the FFT,

(iii) and the number of real multiplications

for the best implementation method. Also, consider the general case where $x[n]$ and $y[n]$ are complex sequences and the FFT of $y[n]$ can be computed in prior. (25 scores)

(a) direct: $3 \times N \times M$ non-sectioned convolution:

$$= 3 \times 1500 \times 250$$

$$= 1.125 \times 10^6$$

$$P \geq M + N - 1 = 1500 + 250 - 1 = 1749 \quad (P \text{ 取 } 2016)$$

$$2 \text{ MUL}_{2016} + 3 \times 2016 = 2 \times 12728 + 3 \times 2016 = 31504 \text{ real MULs}$$

sectioned convolution: $L_0 = 2106$, $P_0 = L_0 + M - 1 = 2106 + 250 - 1 = 2405$

$$\textcircled{1} P = 2048$$

$$S \times (2 \text{ MUL}_{2048} + 3 \times 2048)$$

$$L = P - M + 1 = 1 \times (2 \times 16836 + 3 \times 2048)$$

$$= 1799 = 39816$$

$$S = \left\lceil \frac{1500}{1799} \right\rceil = 1$$

$$\textcircled{2} P = 2304$$

$$S \times (2 \text{ MUL}_{2304} + 3 \times 2304)$$

$$L = P - M + 1 = 2 \times (2 \times 15868 + 3 \times 2304)$$

$$= 2055 = 77296$$

$$S = \left\lceil \frac{2055}{1500} \right\rceil = 2$$

(b) direct: $3 \times N \times M$ non-sectioned convolution:

$$= 3 \times 1500 \times 50$$

$$= 2.25 \times 10^5$$

$$P \geq M + N - 1 = 1500 + 50 - 1 = 1549 \quad (P \text{ 取 } 1680)$$

$$2 \text{ MUL}_{1680} + 3 \times 1680 = 2 \times 10420 + 3 \times 1680 = 25880$$

sectioned convolution: $L_0 = 324$, $P_0 = L_0 + M - 1 = 273$

$$\textcircled{1} P = 288$$

$$S \times (2 \text{ MUL}_{288} + 3 \times 288)$$

$$L = P - M + 1 = 7 \times (2 \times 1160 + 3 \times 288)$$

$$= 239 = 22288$$

$$S = \left\lceil \frac{1500}{239} \right\rceil = 7$$

$$\textcircled{2} P = 252$$

$$S \times (2 \text{ MUL}_{252} + 3 \times 252)$$

$$L = P - M + 1 = 8 \times (2 \times 1024 + 3 \times 252)$$

$$= 203 = 22432$$

$$S = \left\lceil \frac{1500}{203} \right\rceil = 8$$

$$\textcircled{3} P = 240$$

$$S \times (2 \text{ MUL}_{240} + 3 \times 240)$$

$$L = P - M + 1 = 8 \times (2 \times 940 + 3 \times 240)$$

$$= 191 = 20800 \text{ real MULs}$$

$$S = \left\lceil \frac{1500}{191} \right\rceil = 8$$

$$\textcircled{4} P = 204$$

$$S \times (2 \text{ MUL}_{204} + 3 \times 204)$$

$$L = P - M + 1 = 10 \times (2 \times 916 + 3 \times 204)$$

$$= 155 = 25640$$

$$S = \left\lceil \frac{1500}{155} \right\rceil = 10$$

(c) direct $= 3 \times N \times M$ non-sectioned convolution:
 $= 3 \times 1500 \times 10$ $P \geq N + M - 1 = 1500 + 10 - 1 = 1509$ (取 1680)
 $= 4.5 \times 10^4$ $2 \text{MUL}_{1680} + 3 \times 1680 = 2 \times 10920 + 3 \times 1680 = 25880$

sectioned convolution: $L_0 = 42$ $P_0 = L_0 + M - 1 = 51$

① $P = 52$ $S \times (2 \text{MUL}_{52} + 3 \times 52)$ ② $P = 48$ $S \times (2 \text{MUL}_{48} + 3 \times 48)$
 $L = P - M + 1 = 35 \times (2 \times 208 + 3 \times 52)$ $L = P - M + 1 = 39 \times (2 \times 92 + 3 \times 48)$
 $= 43$ $= 20020$ $= 39$ $= 12992$ real MUL_S
 $S = \lceil \frac{1500}{43} \rceil = 35$ $S = \lceil \frac{1500}{39} \rceil = 39$

③ $P = 42$ $S \times (2 \text{MUL}_{42} + 3 \times 42)$
 $L = P - M + 1 = 46 \times (2 \times 124 + 3 \times 42)$
 $= 33$ $= 19204$
 $S = \lceil \frac{1500}{33} \rceil = 46$

(d) direct $3 \times M \times N$ non-sectioned convolution:
 $= 3 \times 1500 \times 2$ $P \geq N + M - 1 = 1500 + 2 - 1 = 1501$ (取 1680)
 $= 9000$ $2 \text{MUL}_{1680} + 3 \times 1680 = 25880$

sectioned convolution: $L_0 = 2$, $P_0 = L_0 + M - 1 = 3$

① $P = 4$ $S \times (2 \text{MUL}_4 + 3 \times 4)$ ② $P = 3$ $S \times (2 \text{MUL}_3 + 3 \times 3)$
 $L = P - M + 1 = 500 \times (2 \times 0 + 3 \times 4)$ $L = P - M + 1 = 750 \times (2 \times 2 + 3 \times 3)$
 $= 3$ $= 6000$ real MUL_S $= 2$ $= 9150$
 $S = \lceil \frac{1500}{3} \rceil = 500$ $S = \lceil \frac{1500}{2} \rceil = 750$

③ $P = 2$ $S \times (2 \text{MUL}_2 + 3 \times 2)$
 $L = P - M + 1 = 1500 \times (2 \times 0 + 6)$
 $= 1$ $= 9000$
 $S = \lceil \frac{1500}{1} \rceil = 1500$

- (5) Are the following applications suitable for the Walsh transform? Why? (a) calculating the linear convolution; (b) stair-like signal analysis. (8 scores)

- (a) Walsh transform 實現的是 logical convolution, $h[n] = \sum_{l=0}^{N-1} f[n \oplus l] g[l]$
 很難用 Walsh transform 做 linear convolution 的計算
 (b) Walsh transform 可以更有效處理短時間劇烈變化, 其他時間平緩,
 像是心電圖, 和 stair-like 訊號

- (6) What is the number of addition operations when we want to implement (a) the 32-point Walsh transform, (b) the 16-point Haar transform? (10 scores)

- (a) $\log_2 32 = 5$ stage, $32 \times 5 = 160$ ADDs
 (b) $16 + 8 + 4 + 2 = 30$ ADDs

- (7) (a) What are the results of CDMA if there are three data $[1 \ 0 \ 1]$, $[1 \ 1 \ 0]$, $[0 \ 1 \ 1]$ and these three data are modulated by the 1st, 4th, and 10th rows of the 16-point Walsh transform? (10 scores)
 (The beginning row is the 1st row).
 (b) In (a), if the 5th and the 15th entries of the CDMA results are missed, can we recover the original data? Why? (5 scores)
 (c) Is it proper to use the Haar transform instead of the Walsh transform for CDMA? Why? (5 scores)

- (a)
 $[1 \ 0 \ 1] \rightarrow [1 \ -1 \ 1]$
 $[1 \ 1 \ 0] \rightarrow [1 \ 1 \ -1]$
 $[0 \ 1 \ 1] \rightarrow [-1 \ 1 \ 1]$

zero crossing

$$V_{16} = \begin{bmatrix} w_8 & w_8 \\ w_8 & -w_8 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \end{bmatrix}$$

0
3
4
7
8
11
12
15

1
2
5
6
9
10
13
14

